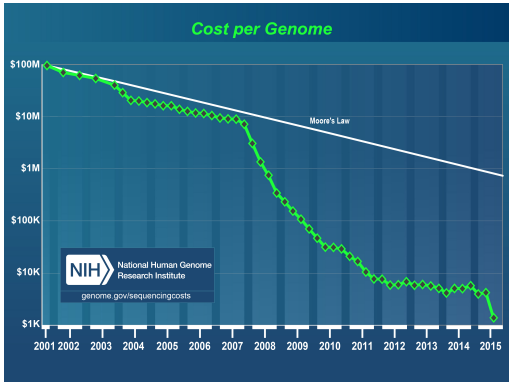
Computational Biology

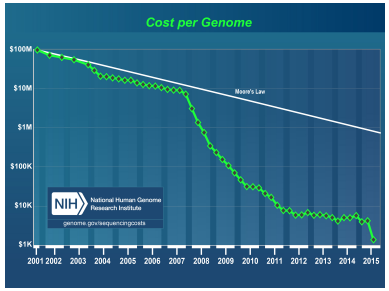
Lecture 3: *Sequencing*



Shotgun, new sequencing technology, nanopores

Outline

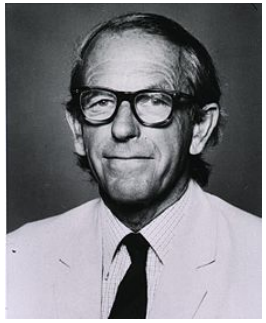
1. **Sanger Sequencing**
2. Next Generation Sequencers
3. Third Generation Sequencers



Sequencing Revolution

- Advances in our ability to sequence DNA lies at the heart of the biological revolution
- Despite the structure of DNA being discovered in 1953, reading DNA sequences took several decades to take off
- Francis Crick had speculated on how DNA might code for proteins
- By removing some number of bases from DNA he was able to show that DNA has a three letter code

Frederick Sanger



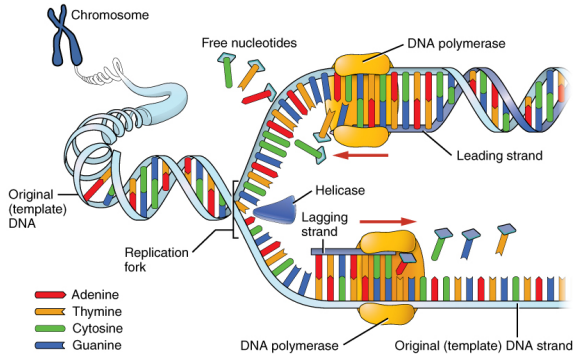
- Frederick Sanger was a British born biochemist
- In 1955 he discovered the protein sequence for Insulin
- This showed that proteins were actually chains
- He won the Nobel prize for chemistry for this in 1958

Sequencing DNA

- Chemically it was difficult because the bases are not that distinctive
- It wasn't until 1965 that the very first strands of RNA were read
- A lot of effort went into learning how to copy DNA and break it up into smaller sequences
- Various techniques were developed in the 1970's to sequence longer lengths of DNA until in 1977 Sanger developed a new method now known as **"Sanger Sequcencing"**

Sanger Sequencing

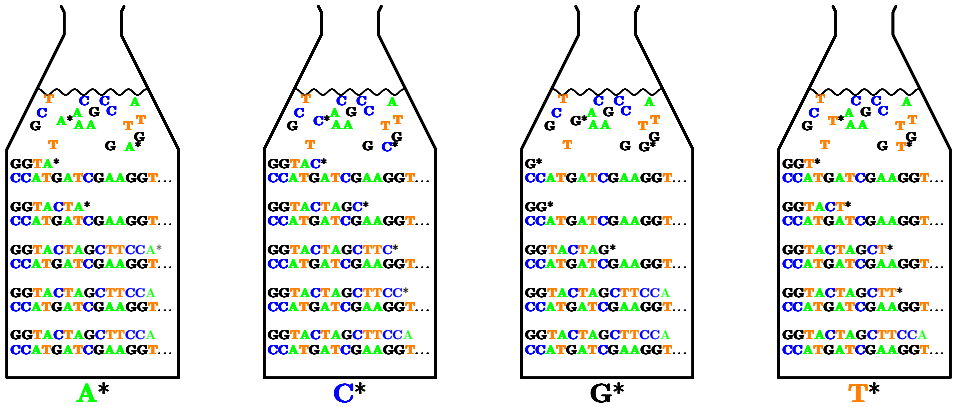
- In Sanger sequencing we use biological mechanisms for copying DNA
- This would add Dideoxynucleoties (ddNTP) to a DNA chain



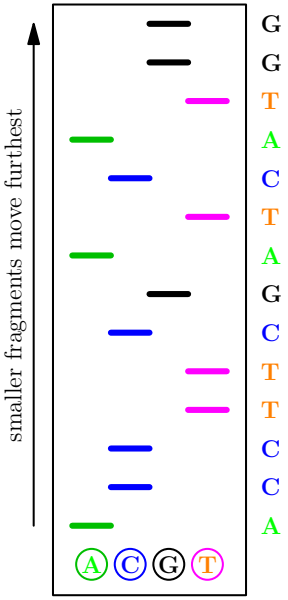
Modified ddNTP

- Modified ddNTP's were developed with a missing $3'$ hydroxyl group which prevented the chain from developing any further
- These were marked (either with radioactive phosphorous or later with a fluorescent marker)
- DNA replication was then carried out with normal ddNTP's and a weak concentration of these modified ddNTP's
- Initially this was done in four containers one for each base (in container A only the base A ddNTP were modified)
- Later different fluorescence were used so everything could be carried out in the same vessel

Sanger Sequencing



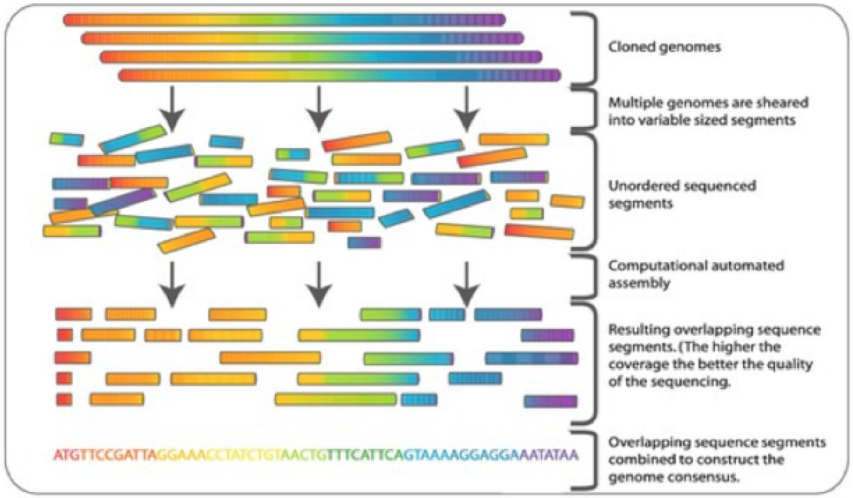
Fractionation



- Each solution is added to a layer of gel
- Using capillary electrophoresis different length DNA fragments are sorted
- From the distance travelled and the colour of the last nucleotide base we can read off the sequence
- This won Frederick Sanger a second Nobel prize awarded in 1980

Shotgun Sequencing

To save careful preparation researcher developed shotgun sequencing



Reconstruction

- Reconstruction looks easy, but the human genome is full of very repetitive sequence
- Sanger sequences only allowed you to sequence "reads" of around 1000bp
- Not clear what the long range ordering is
- For large genomes researchers built "scaffolds" where they knew the position of some genes
- You can sequence both sides of a large fragment so you know the approximate relationship between reads

Human Genome Project

- Sanger sequencing was the technology that delivered the human genome
- Huge advances meant that it was delivered several years earlier than expect
- The cost of sequencing a human genome in 2003 was estimated at around $10-100 million
- The human genome project did not deliver an immediate benefit to medicine that some had anticipated—we still did not and don't understand enough biology

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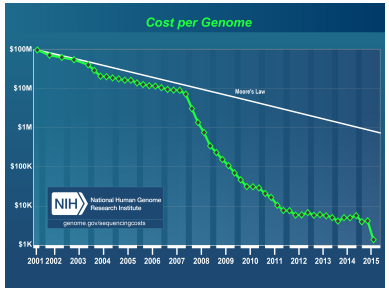


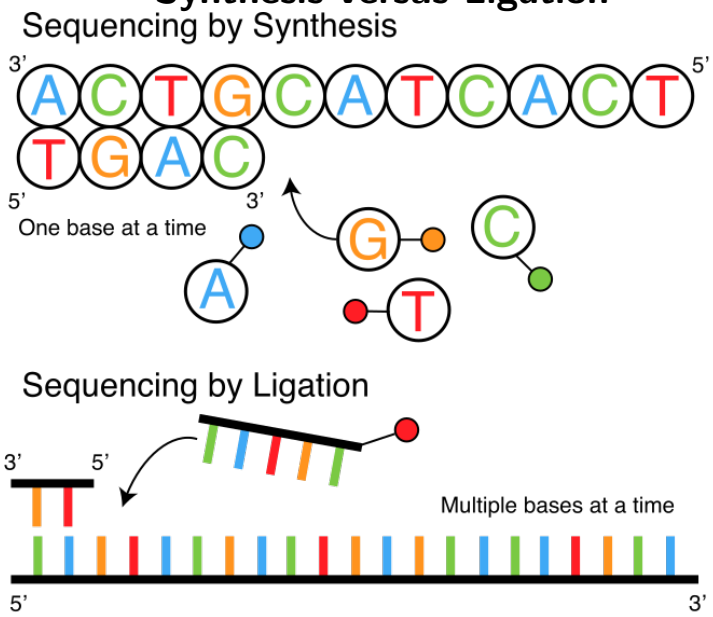
Improving on Sanger

- Although Sanger sequencing saw improves in efficiency in line with Moore's law there was a desire to reduce the cost of sequencing a human genome to $1000 within a decade
- The ideas was to use massive parallelisation
- Many different techniques were tried

Sequencing Techniques

- Sanger sequencing used sequencing-by-synthesis
  - ⋆ Using DNA-polymerase a strand of DNA is replicated
  - ⋆ It turns out this technique that ultimately triumphant
- Another approach is sequencing-by-ligation (or hybridisation)
  - ⋆ Create lots of short sections of DNA (oligonucleotides) that will stick to an unbound stretch of DNA (or RNA)
  - ⋆ Used in microarrays

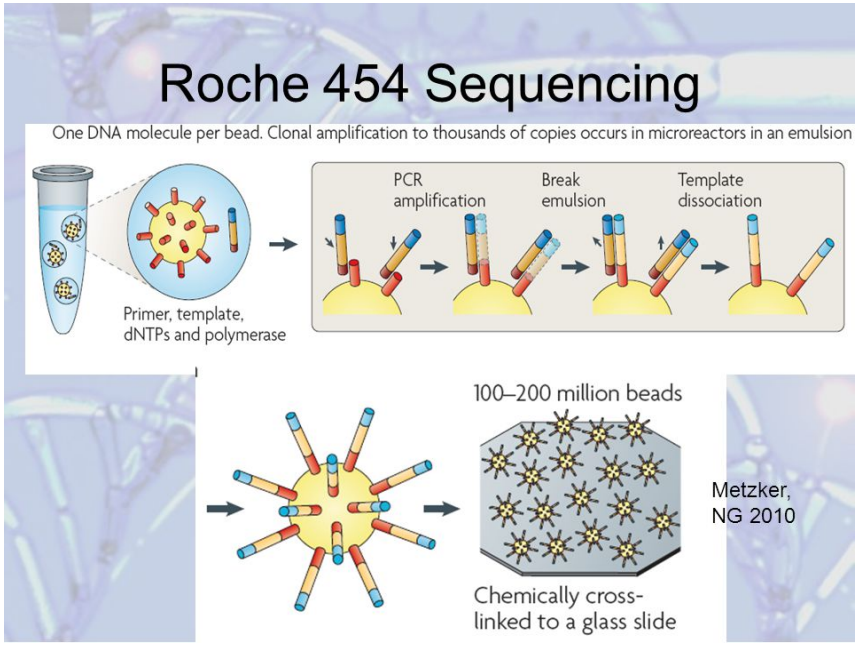
Synthesis versus Ligation



454 Sequencer

- The first high throughput sequencer was the Roche 454 machine
- It amplified fragments of DNA attached to a microbead
- These are contained in micro-wells
- Sequencing-by-synthesis is carried out
- Added one type of base at a time with fluorescent label
- Can have difficulty distinguishing regions with repeated base

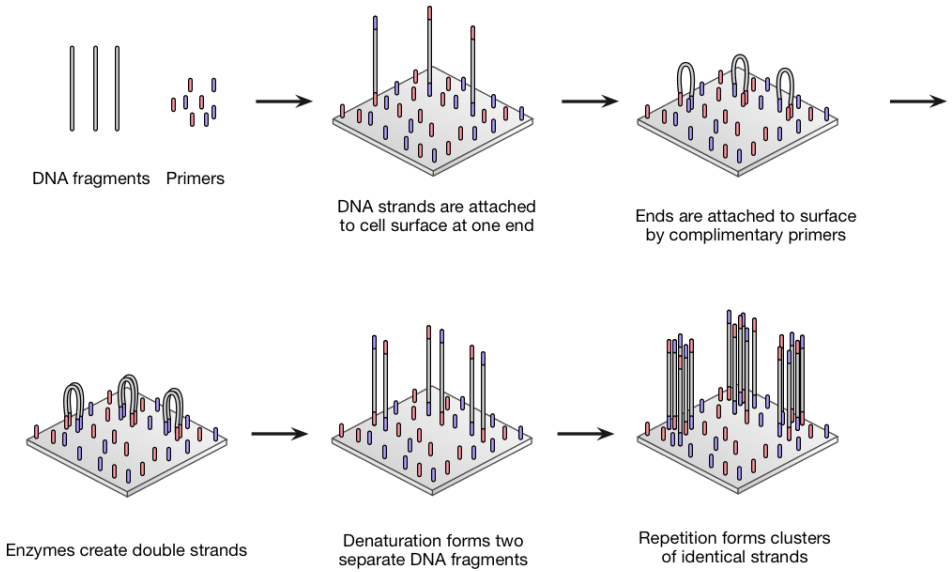
Water in Oil Emulsion PCR



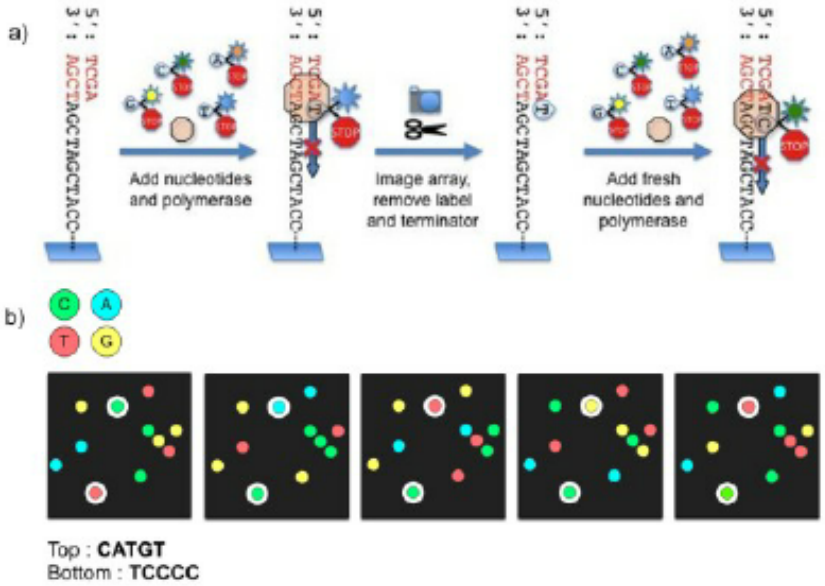
Illumina

- The dominant next generation machines are produced by Illumina (formally Solexa)
- It uses a technique now called bridge amplification to create clusters of identical strands on a 2-D surface (substrate)
- The machine can sequence 100's of million clusters at once
- It then uses sequencing-by-synthesis to read the sequence
- It can also read the other end of the sequence by flipping the sequence
- Try Googling "Illumina sequencing videos"

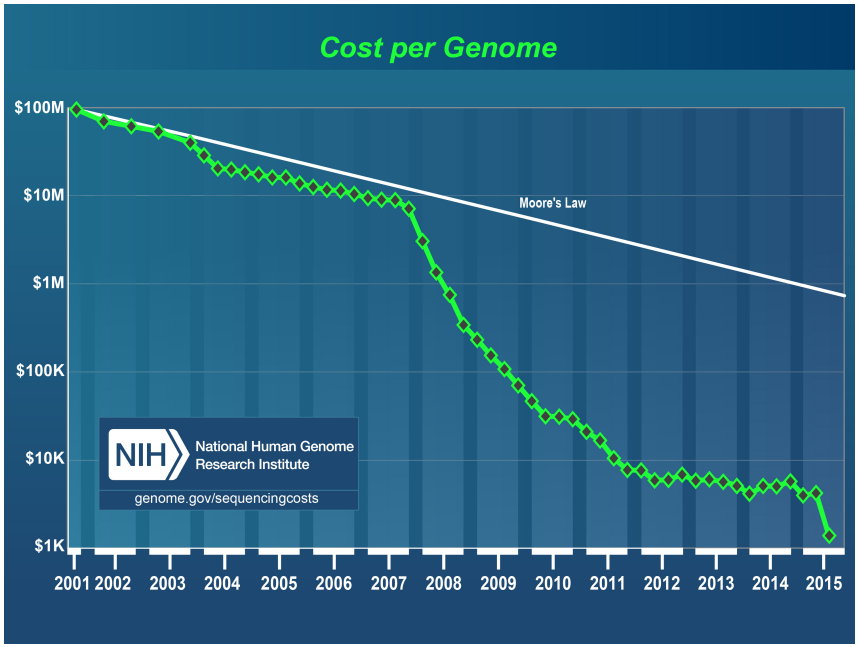
Bridge Amplification



Identifying Sequence



Cost Per Human Genome



100 000 Genome Project

- Currently the NHS and Sanger centre is sequencing around 70 000 human genomes to identify rare genetic diseases
- We can now sequence a human genome within a day
- Because we have already sequenced the human genome we no longer have difficulty reconstructing the sequence

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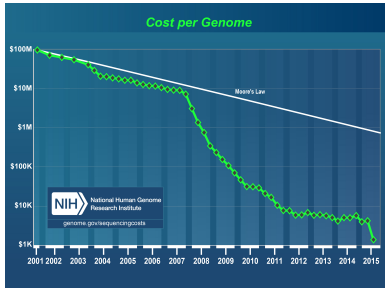
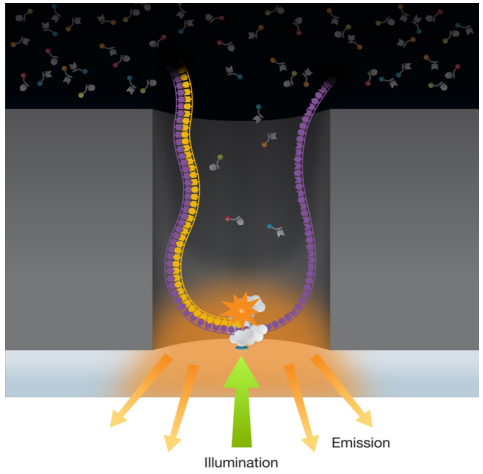


Single Molecule DNA Sequencing

- Having to replicate DNA can bring in errors and biases
- Most of the high-throughput techniques only read short sections (≈ 100 base pairs), which can make reconstruction difficult
- The current research is to create a machine that can read a long single DNA strand

Zero Mode Waveguide

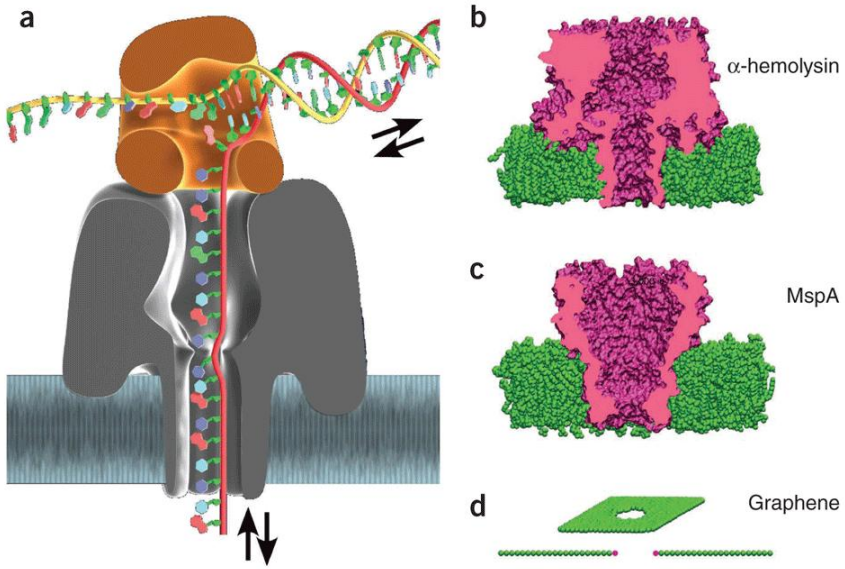


- Uses same fluorescence dNTPs as used in high-throughput processing
- Use micro-well smaller than the wavelength of light
- Able to detect very small signal
- Not high-throughput

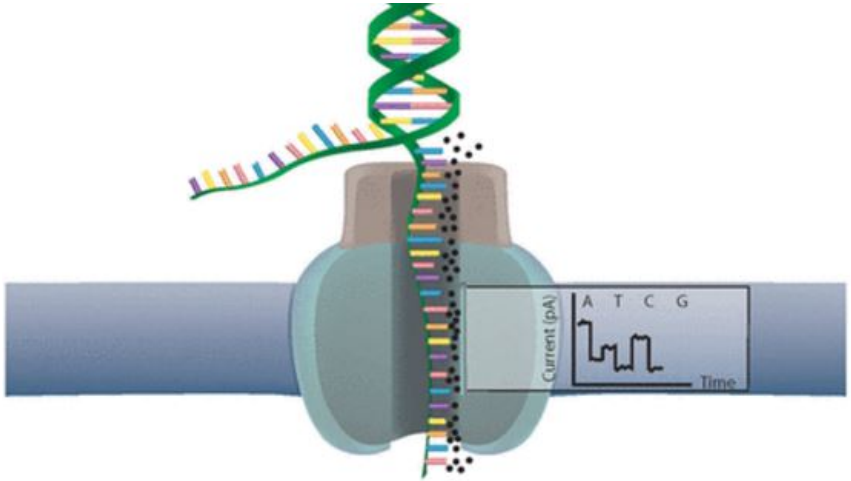
Nanopore Technology

- One of the most anticipated new technologies is to force DNA through a tiny pore and read out the sequence
- Oxford nanopores have been at the forefront of developments
- They have just released a hand-held machine
- Currently the accuracy is not quite good enough

Nanopore



Nanopores



Conclusion

- One of the dramatic revolutions that has occurred in your life times has been the ability to sequence DNA
- Although sequencing proteins predates the sequencing of DNA their has been no such revolution
- The technology builds on the natural tools of the cell (e.g. DNA replicase)
- Computationally the ability to reassemble reads into longer "contigs" has been essential to this revolution